Mohamed Magdy Abdelaziz

Backend Engineer (Java , Springboot)

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Professional Summary

Java Backend Engineer with 3+ years of experience building scalable backend applications using **Java**, **Spring Boot**, and **AI-powered solutions, and AI work flows.** with strong **Spring Cloud**. Strongly grounded in **Clean Architecture**, **SOLID principles**, secure **RESTful APIs**, and **event-driven** systems. Experienced with **SQL/NoSQL** databases and Query optimization, with hands-on experience using **AWS** and **DevOps practices**, observability, **containerization**, and cloud-native technologies.
Experienced with **TDD** and collaborating with international teams to deliver production-ready solutions.

Technical Skills

- **Languages & Frameworks:** Java, Spring Boot, Spring MVC, Spring Data JPA, Hibernate, MyBatis, MapStruct
- **Architecture & Backend:** Microservices Architecture, RESTful API Development, Clean Architecture, Design Patterns, SOLID Principles, Swagger/OpenAPI, Camunda BPM, BPMN
- **AI &Automation:** n8n, AI Integration
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, SQL, Flyway,ACID
- **Testing:** JUnit, Mockito, Unit Testing, TDD
- **Cloud & AI:** AWS, S3, Bedrock, Textract, Rekognition
- **DevOps & Tools:** Docker, Kubernetes, Helm, Maven, CI/CD, GitHub Actions, Jenkins, Terraform, Linux, Git
- **Messaging & Observability:** Apache Kafka, RabbitMQ, OpenTelemetry, Prometheus, Grafana, Tempo, Zipkin
- **Security:** Spring Security, JWT, OAuth2, Keycloak

Professional Experience

Digital Universe 10/2024 – Present
Java Backend Developer

- Engineered scalable backend services for ERP and business management platforms using **Java** and **Spring Boot**, supporting concurrent daily transactions effectively.
- Collaborated with international stakeholders across Saudi Arabia and Asia to deliver **ZATCA-compliant E-Invoicing** integrations, ensuring reliable regulatory compliance.
- Integrated AI-powered solutions via **AWS Bedrock** and **AWS Textract**, automating document processing and reducing manual data entry.
- Optimized complex **SQL**/**MySQL** queries using **MyBatis** and developed enterprise reporting modules with **JasperReports**, improving application performance.

Google Developer Student Clubs (GDSC) 02/2024 – 08/2024
Backend Trainee

- Delivered 3+ **MVC-based** backend applications applying **SOLID**, DRY, and Design Patterns across Java and Spring-based projects.
- Applied **SQL** optimization techniques to relational databases, increasing query efficiency by 25% and reducing response latency.

Projects

EazyBank — Event-Driven Banking Microservices
Spring Boot, Spring Cloud, Apache Kafka, RabbitMQ, Resilience4j, Kubernetes, Helm, Prometheus, Grafana

- Built a **microservices** ecosystem with **Spring Cloud** including Config Server, Eureka, and API Gateway for distributed banking operations.
- Configured event-driven communication using **Apache Kafka** and **RabbitMQ**; employed **Resilience4j** Circuit Breaker and Retry patterns for resilient service communication.
- Orchestrated containerized services using **Kubernetes** YAML manifests and created an umbrella **Helm** chart for multi-environment configuration management.
- Established an observability stack using **Prometheus**, **Grafana**, OpenTelemetry, Tempo, and Zipkin.

E-Commerce Microservices — Cloud-Native Platform
Spring Boot, Spring Cloud, Apache Kafka, MongoDB, PostgreSQL, Keycloak, Docker

- Designed a distributed architecture with independent **microservices** for Product, Order, Inventory, and Notification services using the **Database-per-Service** pattern.
- Engineered asynchronous order processing using **Apache Kafka** and service routing through **Spring Cloud Gateway** and **Eureka**.
- Secured the platform using **Keycloak**, **OAuth2**, and **JWT**; containerized services using **Docker Compose** and automated image builds with **Google Jib**.

Home4U — Enterprise Home Services Marketplace
Spring Boot, Spring Security, JWT, PostgreSQL, Redis, Flyway, Docker, Swagger

- Designed a multi-role marketplace backend connecting customers, engineers, contractors, and suppliers using **Clean Architecture** and **RESTful APIs**.
- Enforced role-based authentication and authorization with **Spring Security** and **JWT**; utilized **Redis** caching and **Flyway** database migrations.
- Containerized the application with **Docker** and documented REST APIs using **OpenAPI (Swagger)**.

Book Social Network — Sharing & Borrowing Platform
Spring Boot, PostgreSQL, WebSockets, Keycloak, OAuth2, GitHub Actions, CI/CD

- Built a social platform with real-time notifications using **WebSockets** and a complete book-borrowing workflow backed by **PostgreSQL**.
- Secured the application using **OAuth2** and **Keycloak** and configured a **CI/CD** pipeline with **GitHub Actions** and a self-hosted Linux runner.

Education

B.Sc. in Computers and Control Systems Engineering 09/2020 – 07/2025
Faculty of Engineering

- **Grade:** Excellent with Honors | **Graduation Project:** Home4U — AI-Powered Marketplace

Languages

- Arabic — Native
- English — (B2)